"I Feel Grateful to the Platform for Letting me Grow": A Study of Older Adults Using Short Video Platforms

LEAVE AUTHORS ANONYMOUS

Short video platforms are growing in popularity, not only among young people but also older adults. However, prior work mainly focuses on younger users, and older adults' use has received far less attention. In this paper, drawing on a qualitative study with 13 older adults who use these platforms regularly for online creation and sharing, we uncover an active use of technology among our participants. Specifically, we highlight how these platforms provide a low-barrier yet fruitful start and a broad encouraging audience for older adults to continue. Short video platforms, then provide a friendly learning curve for older adults to start comfortably and be motivated to learn and improve. As such, an active and meaningful ageing experience is made possible – older adults are motivated to keep practising, engage in meaningful socialization with other users, continually learn and improve themselves, and reflect over their life attitude. Based on these findings, we discuss the insights for how to better engage older adults into active use of technology and related benefits.

CCS Concepts: • **Computer systems organization** → **Embedded systems**; *Redundancy*; Robotics; • **Networks** → Network reliability.

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1 INTRODUCTION

Short video platforms, supporting both asynchronous short video sharing and synchronous live-streaming, have gained increasing popularity recently. These platforms, as a new form of dynamic media, do not only support the watching of videos online but also encourage the sharing of user-generated videos (UGV). While early research suggests the difficulty for streamers to find appropriate topics to stream and the importance of video quality [36], recently, we've seen these platforms rising in popularity, with a variety of topics and activities, such as cultural heritage [46], makeup tutorials, and eating [2].

While younger adults are generally regarded as the main users of these platforms [48, 65, 72], surprisingly, we've noticed that some short video platforms also become very popular among older adults in China. According to a recent report [3], in China, 11% of the users on short video applications are people over 50 years old. While older adults are generally regarded as passive receivers of technology [24], a large number of older adults are found active on short video platforms, and some of them even become Internet celebrities with millions of followers [5]. According to [7], short video applications have become the third most used applications among older adults in China, just following instant messaging and news reading applications, and short video applications is also fastest-growing in the use time among older adults.

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While so many older adults actively engage with short video platforms, past research mainly focused on young people using these video platforms [44, 65] and much less focused on older adults. Researchers found that younger users use short video platforms mainly for entertainment and social needs [45]. They enjoy watching life experience sharing [48], and many newly emerging kind of videos such as life hacking and knowledge sharing [45]. Short video platforms also bring about various new form of interaction, such as digital gifting [48]. Though research on younger users and short video platforms has generated so many interesting findings, the understanding of older adults and these platforms is still limited.

To fill the research gap, we try to explore how and why older adults actively engaged with short video platforms in this study. Specifically, we conducted a qualitative study of older adults actively using short video platforms in China. For our research, we participated on two most popular short video platforms – DouYin (抖音)[1] and KuaiShou (快手)[2], and sought out active older users. Finally, we recruited 13 older participants from these platforms, and we conducted semi-structured interviews with them, to gain a in-depth understanding. Based on the study, we find that short video platforms provide an accessible and encouraging stage for older adults to engage in, which is mostly supported by the low bar for participation and encouraging audience. With the applications, an active use of technology is made possible – older users actively put their performance online, socially interact with like-minded people over the Internet, which then leads to an active and meaningful ageing experience – older users become willing to learn new skills, engage with meaningful socialization with others and even reflect over their life attitude. Based on these findings, this study could provide much insight for design for ageing.

Regarding our research focus, the most important contributions of this study are:
(1) We contribute to the understanding of short video platforms by including the perspective of older adults for the first time.
(2) We discuss the insights for design for ageing based on the case of older adults' active use of short video platforms.

In sections below, after an introduction of the use of short video platforms, we will explain how our work situate in current literature, present our research methods and the findings. Finally we will discuss the results focusing on how to better support older adults' active use of technology, especially for older adults' online creation and sharing.

2 BACKGROUND: SHORT VIDEO PLATFORMS

In this section, we will briefly describe the short video platforms we studied – DouYin and KuaiShou. Both platforms provide short video sharing as well as live-streaming services. Short videos on these platforms range from a few seconds to minutes. To facilitate video making, both also provide many easy-to-use technical tools, such as special filter effects [85], lip-sync songs [73], act-out memes backed by music and other sound clips [56] etc. Everyone, including the newcomers, could easily make professional-like videos with these tools. As for live-streaming, nearly every user could hold live-streaming to make real-time interaction with his/her audience as long as he/she met the requirement (e.g. age) of the platform. During live-streaming, the audience could send "Like", comment and digital gifts to the streamer, which will be directly shown on the screen.

During past few years, short video platforms have gained huge success, which caught the interest of the whole society. It is reported that the myth behind is their in-built recommendation algorithm. Both platforms are all built around the "Explore"/"Recommendation" homepage. The homepage of DouYin is an endless video feed based on its in-built recommendation algorithm, and users could

[1]https://www.douyin.com/, the Chinese counterpart of TikTok.
[2]https://www.kuaishou.com/, the Chinese counterpart of Kwai.

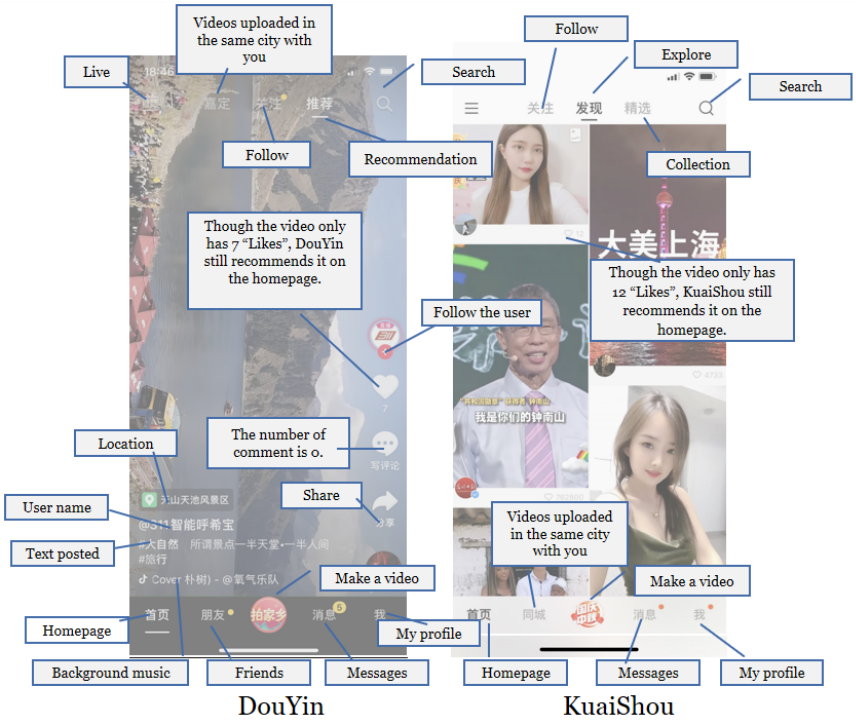

Fig. 1. The homepage of DouYin and KuaiShou. DouYin and KuaiShou are similar in interface design – they all adopt recommendation page as their homepage. Due to the decentralization policy of recommendation algorithm, every video, regardless of its popularity or quality, has a certain chance to be watched, and most content on the platforms are from lay people instead of professionals.

stay to watch or scroll down to watch the next one "pushed" by the platform. Similarly, on KuaiShou, users could click to watch recommended videos or scroll down to view more videos (Figure 1). Due to the design, users generally watch videos "pushed" by the system continuously instead of seeking information by themselves (i.e., "more machine than human" [74]). The system would learn users' preference by their behaviors and navigate users to their favoured contents. Moreover, unlike traditional recommendation algorithms which tend to recommend popular and high-quality content [15], these platforms adopt the decentralization policy – every video, regardless of its popularity or quality, has a guaranteed chance of being watched by others (Figure 1). DouYin, as a typical case, will initiate an original view pool for every posted video [59, 75]. If the video meets specific requirements such as the number of viewers or "Likes", then it will stay in the pool. Otherwise, the algorithm will end pushing it to viewers. KuaiShou is more extreme in decentralization – founded on the core philosophy that everyone is equal and deserves to share their stories with the world [35], KuaiShou's algorithm intentionally steers users away from videos reaching a certain number of hits [88], which is intended to give regular people who otherwise would have trouble getting attention a chance to rise in the rankings.

Surprisingly, though these platforms are generally regarded as the applications used by youngsters [8], many older adults are found to be active on short video platforms. According to [7], on average, users over 50 years old use short video platforms for about 25 hours per week, and it is growing fast.

Many older adults are very popular on the platform and have attracted millions of followers [5]. These phenomena remind us that it is time to pay attention to the older users on these platforms.

3 RELATED WORK

3.1 Older Adults and ICTs

Information and communications technology (ICT) may benefit older adults in terms of social well-being [34], cognitive functioning [84] and physical health [67] etc. For example, the use of ICTs may lead to self-understanding (e.g., increased access to current affairs and health information [49]), social interaction benefits (e.g., increased connectivity and social support [69]), as well as goal-oriented benefits (e.g., shopping,financial management [13]). Empirical studies have also found Internet use could lead to lower perceived life stress[86]. However, the adoption of ICTs among older adults is still not positive, especially for those with less income and education [1]. For example, in America, only 42% of older adults aged 65+ uses smartphone [1]. Therefore, it is highly important to include older adults into the world of technology, especially considering the trend of technology being infrastructural in society [32].

To fill the digital gap, many researchers tried to explain why older adults are resistant to new technologies and build technology acceptance models for older adults [17]. Among these findings, benefits expected of technology and need for technology were reported most as essential factors for older adults to accept technology [55]. This may explain why older adults are willing to use technology for practical goals such as online communication with family members [9] and online shopping [42]. However, reported older adults' use of ICTs is still mainly basic and mundane [66, 78], and older adults' online activities are limited to family, social communication, online shopping,financial management, online gaming and accessing information [78].

Among wide studies on ICTs for older adults, most relevant ones to our work are technologies for online creation and sharing. While older adults are usually described as consumer of digital content [82], recently researchers have found older adults could be motivated to engage in digital content production and sharing if they could derive a meaningful sense from the activity, with both family members [12, 21] and like-minded peers [11, 61]. For sharing with family members, older adults generally see the practicality of using digital media since digital media is almost the only choice if they want to connect with their family members living far away from them [58]. The picture of sharing with like-minded peers is more positive. Brewer et al. found that senior bloggers value the depth of thought and reciprocity; older adults would view their blog as something that will help or be of interest to others [11]. Based on their findings, they further argued that it is important to support older adults' later-life identity development and life enrichment in design for older adults.

Results from research on senior UGV creators are quite mixed. Ryu et al. conducted online surveys with 290 Korean older adults aged 50+, and the study shown that older adults will create UGV if some conditions are satisfied such as perceived benefit, ease of participation and enjoyment [62]. Harley et al. presented a case study of a 79-year-old male video blogger engaging in inter-generational communication through video blogs on YouTube, and they argued that older people could be highly motivated to use communication technologies when social contact is their ultimate aim, and YouTube provides opportunities for reciprocal learning as an online community [30]. However, it should be admitted that on YouTube most older adults only watch videos that are meaningful to them or suggested by people they know very well, and they don't upload videos because they perceive no benefit in it, and online comments are also seldom read nor made [63].

Besides, researchers in HCI also study senior users/non-users of social network sites (SNS) to understand older adults' online creation and sharing behavior. According to [1], only 34% of

American older adults aged 65+ uses social media. Results show that although older adults use SNSs to connect with family and friends [33] and with people who have shared interests [41], they usually only keep their original social connections, with a small networking group [37], and they are less likely than younger users to actively contribute content on these sites [38]. Older adults also seldom leave comments or post status updates [33] because sharing on SNS are perceived as superficial and meaningless, and privacy issue is still a general concern [41]. This is highly consistent with senior non-users' initial perceptions of social media. In [87], Xie et al. discussed with senior non-users of social media, and they found that elderly non-users' initial perceptions toward social media are unanimous and overwhelmingly negative – they usually associated social media sites with disruptive teenage behavior such as cyber-bullying and gossip. However, also in this prior work, after improving their knowledge of social media, older adults could hold realistic conception of social media by realizing the technology's strengths and limitations, and they can be inspired to generate content when content generated is tightly connected to their life.

Research above suggests that, while older adults may generally be resistant to and hold concerns toward new digital technologies, their concerns could possibly be overcome through proper educational strategies that promote personal relevance, and they have the ability to actively use technology for their diverse needs such as online creation and sharing.

3.2 Short Video Platforms

While short video platforms caught the interest of many researchers, most focused on the younger population [45, 46, 48, 80] and research on older adults' use of these platforms is still limited. Specifically, research on short video platforms could be divided into works on short video sharing and live-streaming. In terms of short video sharing, Lu et al. found that Chinese users mainly used short video platforms for entertainment and sociality [45]. In this work, they also identified several unique genres of content shared on short video platforms, including "Positive Energy" ("正能量"), knowledge sharing, life hacks etc. However, apart from this study, research on short video sharing is quite limited, since short video platforms are newly emerging trends in China, and there is no equivalent in other countries. Prior work has mainly investigated public video sharing platforms such as YouTube, Vine, and Snapchat. A survey of online videos conducted by Pew Research in 2013 found that adult content producers mostly post videos of family and friends doing everyday things, themselves or other people behaving in funny ways, and events they attend [16]. While adult mainly regarded online video sharing as an archive, Yarosh et al. found that children and teenagers treat online video sharing as a stage to perform, tell stories, and express their opinions and identities in a performative way [89].

As for research on live-streaming, Lu et al. identified that in China, most viewed content during live-streaming is personal experience [48], which is unlike North American where live-streams were mostly conversational [72]. Tang et al. reported that most of the streamers on Meerkat and Periscope are motivated primarily by branding and self-presentation purposes [72]. Vicarious experiencing is also reported as a main motivation for viewers according to research on eating-related [2] and outdoor live-streaming [47]. Viewer engagement is another primary issue which many researchers have focused on. Research on gaming related live-streaming found that content structure is highly important for viewer engagement [68]. Decisions for digital gifts sending is also reported to contribute to viewer engagement [90].

Though short video platforms provide rich content and interaction to users, what accompanies may be the dark side. For example, the monetary incentive on these platforms was reported to discourage streamers from streaming when they treat their levels of gift income as metrics of their success as streamers [80]. Also, to attract viewers, streamers may exaggerate their performance and hurt their health [2]. Surprisingly, users perceived little negative content on Chinese short

video platforms, possibly because trolls are often blocked by the streamers or their moderators, and inappropriate content is often cut off by the official platform regulators [48]. Some reported that some videos were so negative for emotions that made users hopeless about real life, and even brought contagion of negative emotions; Concerns about addiction to short video platforms, and its potential impact on younger generations were also reported [45]. The privacy issue is still a general concern, as some noted that the system could recommend very relevant ads to them [45].

However, what remains unclear is how older adults make use of these newly emerged platforms. In most past work, younger adults were assumed to be the main users of short video platforms. Though demographic survey confirmed this assumption [48], older adults are also active on the platforms and should not be ignored. To our knowledge, until now there is still no comprehensive study on older users of short video platforms. Our work then bridge this research gap by exploring this large group of users for the first time.

4 METHODS

Prior Search and Observation. In this work, we define older adults as those who are over the legal retirement age (50 for women, 60 for men in China [83]), and we mainly focus on the most popular short video platforms, KuaiShou and DouYin. To gain a general picture of what older adults share and how they interact with each other on the platform, we first observed several older adults' accounts, which we found by using age-related keywords such as "50 Hou", which means "born in the 50s" in Chinese. After finding the seed users, the platform recommended more similar users to us, thanks to the recommendation algorithm. We followed some of them and engaged in conversations with them during live-streaming or through direct messages, to get more understanding of and more direct experience of their online behaviours. We then started to recruit older users for interviews, for a deeper understanding of how they used short video platforms in their lives and what roles these platforms played for them.

Recruitment. We sent recruitment messages to hundreds of candidates on these platforms. Out of ethical concerns, in the direct message, we explicitly state our intention and personal background to older users, and we talked with them only after their explicit consent. We recruited the most participants from Kuaishou as the response rate on Kuishou was the highest, probably because of the particular persona of Kuaishou, i.e., most of the users are from countryside [23, 46] and thus may feel isolated and appreciate the platform as a way to connect with others over the Internet. In the end, we had 13 senior users willing to participate in the interviews, as shown in Table 1. All participants were checked by us to make sure that they were older adults and that they used their accounts by themselves instead of relying on help from their family members or other people. All participants were directly recruited from KuaiShou and DouYin, except for P12 who was introduced by P10. We reached saturation until we felt that data collected were highly consistent and our research questions were covered.

Interviews. We conducted semi-structured interviews with these 14 participants through online chat, audio calls or in-person meetings, based on the participants' preferences. Each interview lasted from 40 to 60 minutes and questions included how they got to know the platform, when and how they used the platform, how they made videos, and how they interacted with others online and offline. During the interviews, we tried to seek specific information about their adoption and use of the platform. Since all our participants are Chinese, all the interviews were conducted in Chinese. All the interviews were audio-recorded, upon the explicit permission of the participants to avoid ethical problems, and notes were taken during each interview.

Data Analysis. We adopted the grounded-theory approach [28] to qualitative analysis , which involves iteratively reviewing and labeling the emerging codes. Two Mandarin-speaking authors coded the transcripts independently and discussed the results to reach a consensus. Through open

ID	Sex	Age	Occu-pation	Edu-cation	Living with	Loca-tion	Main Content	Years of Use	# of Following/ Followers	Plat-form	Live-streaming	Other
P1	F	54	social worker	medium	husband, son with mental illness	city	singing, dancing	1	4683/9048 (0.518)	D		
P2	F	53	retiree	low	parents	rural area	cooking, singing	1.5	844/1022 (0.826)	K	☺	
P3	M	60	part-time security guard	low	wife	city	harmonica playing	0.5	1484/880 (1.686)	K	☺	
P4	M	62	re-employed teacher	high	wife	small town	Erhu playing	2	1470/2778 (0.529)	K	✓	
P5	M	69	farmer	low	wife	rural area	Erhu playing	0.5	1500/996 (1.507)	K	✓	
P6	M	56	retiree	low	wife	rural area	personal life, comic dialogue, chatting	3	753/13,000 (0.058)	K	✓	physi-cally chall-enged
P7	F	60	farmer	low	husband	rural area	singing	1.5	1362/981 (1.388)	K		
P8	M	59	retiree	medium	wife	rural area	Chinese painting	1	1034/9680 (0.107)	K	✓	
P9	F	55	dormitory care taker	medium	husband, son physically challen-ged	city	singing, dancing	1	4574/4312 (1.061)	D		
P10	M	62	retiree	low	wife	small town	Erhu playing	<0.5	1479/750 (1.972)	K	✓	
P11	M	63	retiree	low	wife	small town	Chinese opera	>0.5	1500/1151 (1.303)	K	✓	
P12	F	55	dormitory care taker	medium	husband	city	singing, personal life	1	4900/4900 (1.0)	D	☺	
P13	M	67	grocery owner	medium	wife	small town	Erhu playing, travelling	0.5	721/1281 (0.563)	K	☺	

Table 1. Summary of the participants interviewed. 'K' denotes "KuaiShou" and 'D' denotes "DouYin". ✓ denotes current streamer and ☺ denotes willingness for holding live-streaming.

and axial coding of interview transcripts [10], themes started to merge and brought us back to the transcripts to find more data for them. We used Xmind[3] to arrange the notes into a hierarchy of themes. After several iterations of analysis, a consensus on themes was reached. All the quotes used in the paper were translated into English by the lead author and checked by another member of the research team. All the data is anonymized to protect the privacy of our participants.

Ethical Considerations. Our study was reviewed and approved by the Research Department of the home university of authors. Out of ethical considerations, we explicitly state our intention

[3]https://www.xmind.net/, a popular mind mapping tool.

and background information before every talk. All data collected are upon the permission of our participants and will be anonymized in this paper.

5 FINDINGS

In this section, we will report 1) older adults' general use of short video platform, 2) the reason why they are so passionate about short video platforms, and 3) what the use of short video platforms did for older adults' retired life.

5.1 Overview of Older Adults' Use of Short Video Platforms

The use of short video platforms had become a daily routine for all our participants. All reported that they spent more than 3 hours on these platforms every day. Most of them had retired or were not busy at work, and usually, their adult children were not around; as such, playing on these platforms had become their favourite way to spend their free time. Regarding the general concern about overuse, all participants reported that the use of short video platforms did not bother their ordinary lives since it just replaced their traditional entertainment. That is, they spent the time they used to spend on traditional entertainment, such as TV, Mahjong[4], karaoke and public square dancing (广场舞), on the platform.

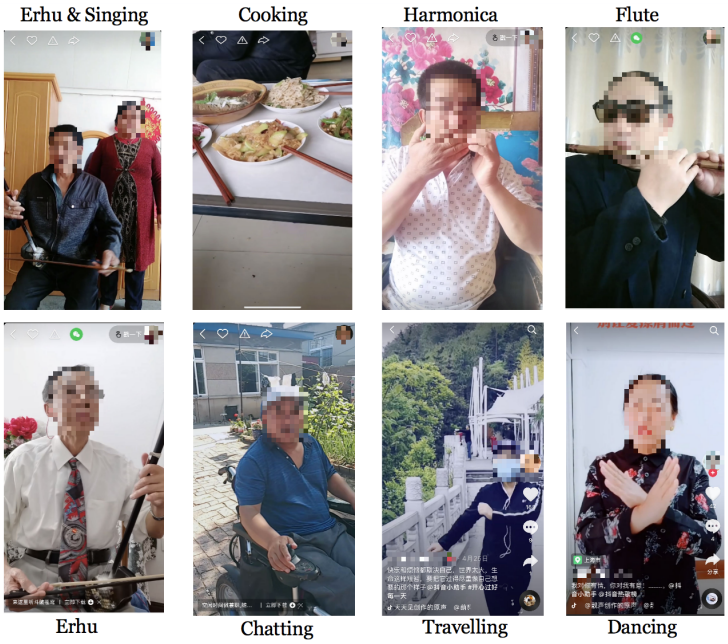

Fig. 2. We identified three main categories of content in older users' shared videos – Conversation, Personal Life and Performance. Surprisingly, among these kinds of videos, older adults are extremely passionate about presenting their performance on the platform.

Short Video Sharing. All participants make their own videos regularly, and they posted about four videos per week. We identified three main categories for their videos – *performance* (e.g. musical instrument playing, singing and dancing), *personal life* (e.g. travelling, cooking), and *conversation* (e.g. comic dialogues, chatting), as shown by Figure 2. Most participants stuck to one category of

[4]A game of Chinese origin played by four persons.

content to stream (Table 1). Surprisingly, though personal life is the most accessible content, the most-streamed category of content is *performance*. The reason reported was that they regarded short video platforms as a platform for fun and communication. When asked why they did not share personal life, most said showing their personal life was boring and meaningless, which nobody would want to watch.

Live-Streaming. Although most participants shown willingness for live-streaming, only 6 of them were current streamers. The category of content in live-streaming aligned well with their videos shared. *Performance* and *chatting* are most live-streamed. Compared with short videos, live-streaming offered much more direct and richer interaction between the streamer and audience, which was reported as the main motivation for live-streaming. P5 felt pleasant that he could play the favourite songs ordered by his audience during live-streaming. Similarly, P6 said he could talk directly to his audience through live-streaming, and he also regarded live-streaming as a good way to know his audience better so that he could decide whether to further interact with them or not. On the other hand, live-streaming requires much more physical capacity and time than short video sharing, which limits the frequency and time of live-streaming for older adults. According to the older streamers, generally each time before live-streaming, they would practice his/her performance for about one hour to ensure the quality of their performance. P5 once hold live-streaming every day, but due to the cost of energy, he soon decreased the frequency. P6 also reported that he used to hold live-streaming three times a day and felt uncomfortable.

Interestingly, though monetary incentive is the primary motivation for younger streamers [80], most participants put much more value in the interaction with others than money, although nearly all streamers we interviewed received digital gifts during live-streaming. This was also true for non-streamers. P1 told us she received an invitation from a business company to be a professional streamer, but she refused to do so because live-streaming cost time and energy and she was reluctant to make money through live-streaming, which, she regarded as the tool for communication.

5.2 A Social-technical System with a Low Entry Level and Encouraging Continuation

5.2.1 Start with Content of Various Varieties and Qualities. We found the main advantage of these short video platforms for the start of use is that they provided fruitful content and a low entry bar, which easily caught the interest of our older adults and offered them much confidence in the use. Specifically, short video platforms automatically provided a large amount of content for watch at homepage (Figure 1), which caught users' interest at first impression and largely facilitated information seeking. Moreover, based on the decentralization policy of recommendation, these content are inclusive in both variety and quality, which adds much fun for our older adults and further lowers the bar for participation.

All the participants reported that they were very impressed with the rich content on the homepage when they first opened the application, and they were soon attracted by recommended videos on the homepage. Notably, though interested, some with little experience in smart phone use felt resistant to using the application out of assumed fear of technology, but they also decided to have a try with the encouragement of their relatives, and they also felt comfortable with the further use (P8, P12). All participants then relied on the recommendations given by the system to get content to watch and seldom sought for specific information, just as the younger users since videos were continuously "pushed" to them by the recommendation algorithms [45]. Some with specific interests would seek for related videos by search when they noticed users sharing common interests with them, and the system soon recommended much related content to them and helped them find similar users.

Notably, featured by decentralization policy, the recommendation algorithm contributes to the diversity in content variety and makes our older adults feel fruitful. In essence, content of any

category would be recommended on the homepage. P6 used to use YY, another popular live-streaming platform in China but mainly promoting content featured by singing and dancing. However, P6 soon turned to KuaiShou. He said, "People do all kind of things on KuaiShou, including music playing, make-up and travelling, not limited to singing and dancing. In this way, YY is a little bit monotonous and boring." P7, P9 and P12 used to use WeSing[5], but soon gave up when they started using DouYin or KuaiShou. As P12 explained, *"WeSing is only for singing. However, on KuaiShou, you can show all your talents."*

Moreover, thanks to the decentralization policy, videos recommended on the platform were of various qualities and most were from laypeople, not from professionals, thus offering older adults a lot of confidence and encouraging them to participate. P3 was a typical case. He said, *"I am very passionate about video streaming since I found my performance was not so bad compared with the recommended videos."* However, when he found a professional player on the platform after a period of use, his passion for performing reduced a little, *"I was shy to show my performance since the professional player played much better than me."*

As another similar popular application driven by user-generated content (UGC) in China, Weibo[6] was often mentioned for comparison by our participants. Though also supporting video sharing and live-streaming, all participants who knew Weibo felt resistant to use (P3,P4), and those who used Weibo before all gave up at last (P1,P10,P11). Besides some reported difficulty of typing (P3) and expressing with words (P3,P11), reasons reported include that content on Weibo is far away from their life and they could learn nothing from it (P11). Indeed, Weibo is celebrity-oriented and mainly promotes high-quality content from recognized accounts [92]. P11 said:

> *"The recommended posts on Weibo are far away from my life, such as variety shows and pop stars. They are too perfect, and I can learn nothing from those things. However, content on DouYin is close to my life, and I can learn a lot from these videos."*

On the other hand, what possibly accompanies with the increasing diversity of content is the higher risk of negative content. Surprisingly, overall our older participants were left with a good impression on the content streamed on the platforms. Most told us they seldom watched annoying content during the use, possibly because Chinese state authorities have already strictly regulated contents on short video platforms to respond to critics in April 2018 [4]. Most reported negative content by our participants is erotic videos. P6 and P7 all said they sometimes watched scantily clad girls and felt discomfort, and they wanted the platform to make a rigorous regulation on these users and content. P7 told us this way,

> *"I watched some girls showing their tits during live-streaming possibly for money. This is not good, and I will leave the live-streaming room immediately."*

5.2.2 Stay for a Broad and Encouraging Audience. While the low entry level and attractiveness of content gets older adults started, it is the people there that keep them continuing to use the platform. Here, the decentralization policy not only works for the promotion of diversity in content but also succeeds in bringing a large number of audience to every user by recommending their works to others. By giving every user a certain degree of attention, every performer on the platform was able to be treated as the centre on his/her stage. Moreover, such a decentralization policy largely promotes the interaction on the platform and fosters an encouraging environment – every user's work deserves a recommendation and being "Liked". P3 noted,

> *"I never thought of the possibility that others will thumb up for my work. Even professional players sent 'Likes' to me. I felt extremely good and kept posting videos since then."*

[5]https://kg.qq.com/, a mobile application for sharing singing.
[6]https://weibo.com/, a Chinese equivalent of Twitter, also supporting short video sharing and live-streaming.

Below, we will illustrate about the two main important characteristics of the audience on short video platforms for older adults – broad and encouraging.

A Broad Audience. We found that short video platforms worked as a stage by providing interesting peers who were either like-minded performers or audience members. Many of our participants expressed the gratitude to reach more people, not constrained by their social networks or geographical limits [51]. Many reported never having known so many other people shared their interests before using the application, and they were extremely surprised at this. As P4 noted,

> *"I am a member of a band in my town, but I never knew that there were so many Erhu[7] lovers in the world... I want to meet all my friends on KuaiShou in reality...But they live too far away from me, including Guangzhou, Hunan, and other far-away places."*

A stage like this became especially meaningful for those living in rural areas or small towns, as they had even less of a chance to meet other people and perform for others in real life. All the participants from rural areas responded that there were either no or very few places for them to go and have fun in real life. P8, who lived in a rural area, said,

> *"There is not much for fun for older adults living in rural areas. I learned Chinese painting before, but nobody around can paint with me. I saw so many painting lovers on KuaiShou, so I started posting videos of my Chinese paintings and am having a lot of fun right now."*

P5 used to go to the square in a nearby town to perform Erhu, but couldn't go there often since the town is far away from his home. But with KuaiShou, *"Now I play Erhu for others every day at home, and the performance can be seen by people nationwide instead of only by people in the town."*

The isolation described by P8, P5, and P13 is getting worse as more and more people, especially the young, are leaving rural areas to work in cities, making it even harder for older people to meet people who share similar interests offline. P7 said that the only entertainment she had before using KuaiShou was gossiping with others, which was boring for her. Although she once organized a dancing group, it soon failed because many people left. Meanwhile, for those who are physically disabled and cannot go out often (e.g. P6), using a short video platform was a way to get to know the world without going out.

An Encouraging Audience. Interestingly, quick feedback from followers are extremely appreciated by all our participants, and they took all the feedback they received including 'Likes', follows and comments seriously and viewed the feedback as acts of encouragement and respect. In return, they also actively gave feedback to others to show their encouragement and support. Overall, we found that the feedback participants received from the platform was very positive.

Many reported that the attention and positive feedback they received from their followers was the main reason they continued using the platform (P1, P3, P4, P10, P11, P13), and this could largely be attributed to the recommendation policy of the platform. Here, Weibo was again used for comparison, where attention was hard to achieve. P1, P4 and P10 all reported the difficulty of getting feedback on Weibo. Take P4 and P1 for example,

> *"I used Weibo before, but I soon gave it up. I think this may have been because of the lack of interaction with others. On KuaiShou, once you post your video, you will receive a 'Like' very soon. You see, I have 3000 followers right now." (P4)*

> *"I continue playing because of the large number of followers on DouYin. I used Weibo before, but no one viewed my post, so I stopped." (P1)*

Moreover, we found that feedback was often given in a reciprocal manner (P1, P2, P3, P5, P7, P9, P11, P12, P13). Many reported that to be polite, they would follow others back if others followed them. As some comments noted,

[7] A two-stringed bowed musical instrument.

> *"If others follow me, I will also follow them. KuaiShou is a platform for communication, so we should show our support to others." (P11)*
>
> *"I follow others only if they follow me. Others' following shows respect. My education level is low, but I know the importance of respect." (P7)*

We found several participants who very actively and extensively used the 'Like' function. It was common that our participants actively leveraged the "Like" function to encourage others. Some used 'Like' to encourage others instead of showing whether they liked the video or not. P5, who 'Liked' every video he watched, was an extreme example:

> *"I will send a 'Like' to every video I watch, regardless of its quality. I want to encourage others, even if they do not do a good job. Unlike some people who only send 'Likes' to high-quality videos, I want to show my support to every user."*

Similarly, P9 also sent 'Likes' to videos as an encouragement rather than as a reflection of quality. As an active performer herself, she could appreciate how hard it was to produce good work, so she would 'Like' a video even if it wasn't so good. She said, *"I practice many times before posting, so I know that it is hard to take a good one. Even if the video looks not good, I know the user put a lot of effort into it, so I will send 'Like' to him/her."*

The reciprocity is not limited to light support such as "Like" and comments. Some said that they would curate content for their followers. P5 used to play Erhu exclusively, however, after many others followed him, he started to share the view of his hometown and played harmonica, because he wanted his followers to have fun. To P11, it was his followers who helped him overcome all his difficulties using the platform at the beginning. He noted,

> *"Though it was not easy for me to use the smartphone, I continued because of my followers. To bring happiness to the people supporting me, I will try my best to meet their expectation, and the difficulty is not a big deal. If I haven't posted my videos for several days, my followers will comment that they want to see my new work. To show my appreciation for their support, I will prepare for my new videos carefully."*

However, when asked whether they were willing to make deeper communication or form long-term relationship on the platform, most gave the negative answer. Fraud online was still a general concern. For example, P1 said, *"Usually I don't talk too much with others online. There are some bad people online, who will try to cheat you."* Some chose to communicate with others deeper when the interaction was perceived as meaningful such as advice exchanging. As a typical case, P3 would send direct messages to those seemingly professional players because he wanted their advice for music playing. P6 only formed a long-term relationship with those he is familiar with during live-streaming, but he still had some negative experiences toward this,

> *"I will talk with those who often come to talk with me during live-streaming because I am familiar with them. I can tell you that most people are kind. However, one time someone asked me for money to pay for his mobile phone bills. I chose to trust him at that time. But soon I found he was a fraud."*

Despite this negative experience, he still chose to trust others because *"Most people are good."*

5.3 Towards a Meaningful Ageing Experience

We found that, for older adults, the active engagement on these video platforms was perceived to make their retired life substantial and meaningful, and this marks the main value of the use of short video platforms. Empowered by other users on the platform, older adults actively learn various skills to improve their video and themselves, and the social support received on the platform also makes older adults feel meaningful in their retired life.

5.3.1 Ageing with Learning: Older Adults as Active Learners. We found that one of the main influence short video platforms made on older adults is that these platforms offered rich resources as well as extra incentives for our participants to learn and to continually improve, which turn older adults from passive viewers to active learners.

Learning and Improving. Many older adults regarded video quality as the determining factor of their attractiveness to others, which pushed them to keep practising and learning to improve. For example, P7 noted, *"I want more followers, which needs me to practice more and improve my Erhu skills."* Thus, the learning mechanism of short video platforms goes beyond mere information sharing and exchange as on traditional online communities [14], since the platform provided extra incentives for older adults to practise more and improve themselves.

Besides hobby and performance-related skills, older adults also actively learn to acquire other skills to improve video quality and watching experience. For instance, some female participants were motivated to make themselves look better by makeup and appropriate dressing, which further changed their lifestyle. Two of our female participants described it this way,

> *"I wanted to learn to do makeup when I was young; however, people in rural areas regarded women doing makeup as strange. Now, I am using lipsticks my daughter left for me, and I feel better now." (P7)*
>
> *"I realized the importance of looking nice after posting my videos. In reality, I've also paid more attention to how I look now. I didn't do makeup before, but I will pick clothes for myself and do makeup now. Looking nice would make me feel energetic." (P1)*

As noted in these two quotes, they brought what they acquired from the platform into their everyday life, and this empowered them a lot.

Besides, by engaging in conversations with people from across the country on the platform, many were able to acquire skills for social integration. For instance, in the offline world, some older adults primarily spoke their region's dialect and did not speak Mandarin very well. Several participants noted that by actively performing on the platform, they were able to improve their Mandarin (P5, P13). P5 spoke Mandarin in videos of himself, and said,

> *"I spoke dialect after retirement. For communicating with others effectively, I started speaking Mandarin again, and my Mandarin has improved a lot since then."*

Video-making itself was also a subject for learning. Our participants reported learning related skills such as adding captions (P3), background music (P10 and P11), special effects (P9) among others. Usually, when they watched a nice video, possibly using some special effect filterings, and they did not know how to make a similar one, they would send direct messages to the author or ask their family members or offline friends. P11 even joined a QQ[8] group chat where people would share guidelines of special video making, to learn related skills.

Moreover, many of our participants were very interested in learning about live-streaming, though it usually takes more complex devices and skills other than video-making and posting as mentioned. The participants who live-streamed (P4, P5, P6, P10) told us that they had been nervous about expressing themselves the first time they did it. However, with the encouragement from the audience and growth of experience, they now felt more comfortable. P4 and P5 shared their similar experience with live-streaming,

> *"I was extremely nervous when I live-streamed for the first time. I thought I was not good enough to perform in real-time. However, with the encouragement of my followers and practice, I am not nervous anymore. Before every live-streaming, I will practice from half an hour to one hour." (P4)*

[8]A popular application for instant message sending in China, similar with WeChat.

> *"I think I usually play Erhu fairly well. Nevertheless, once I stood on the stage of the live-streaming, my performance was terrible at first. I started understanding the Chinese idiom that one minute on stage needs ten years' practice. Live-streaming is a good way to practice yourself. I am grateful to the platform for letting me grow." (P5)*

As shown, doing live-streaming is more challenging and demanding than video-making, but at the same time, it also more effectively pushes people to practice and improve one's performance.

Learning Together: Learning-oriented Socialization. As our participants incorporated video making into their daily life, learning and teaching have also become important motivations to the social interactions of the older adults, both online and offline. Many put a lot of effort into teaching others how to use the platforms, such as P11 and P9,

> *"I made friends with another Chinese opera lover on KuaiShou. We knew each other because she asked me for advice on opera playing. Besides skills in opera playing, she also asked me questions like how to add the caption." (P12)*
>
> *"I would repost my DouYin videos to my WeChat[9] groups. Many of my friends would ask me whether I could teach them, and I would be pleased to offer help. Through my recommendations, many of my retired friends started using DouYin!" (P9)*

Notably, such kind of learning-oriented socialization is hugely appreciated by older adults with specific interests, since it is hard to find like-minded peers, as we mentioned. Among our participants we interviewed, music lovers all actively engaged in interaction with like-minded peers for communication and advice changing. For example, P4 would send music scores to other music lovers on Kuaishou and talk about the experience in Erhu playing with others online. Similarly, P3 was eager for the advice from seemingly professional players on the platform, and P5 was happy to talk with Erhu lovers of different age groups.

The platforms were also able to bridge the gap between generations through learning and teaching. P1 noted that DouYin created a new topic of conversation across generations at work. She said, *"My colleagues also play DouYin. Some younger women in my office also come to ask me for advice on how to look better in the videos."* This is also true for family members. P13 described, *" When I faced difficulties in using KuaiShou, I would ask my family members for help. I think this helps to promote our communication. We used to have almost nothing to talk about."*

Specifically, many leveraged the platforms' live-streaming as a way to teach and learn about live-streaming. P6 used live-streaming to make new friends online by teaching others how to do live-streaming. He reported,

> *"There is a guy who comes from Shanxi. He even visited my home. He enjoyed my live-streaming and asked me for advice on how to live-stream. He often comes to watch my live shows and chat with me. He shows respect to me and regards me as his teacher. He will ask me anything if he faces difficulties during live-streaming."*

Learning, therefore, enriches the ageing experience of our older participants. Before using the applications, some reported that they only gossiped or talked about nonsense with others in real life (e.g.P9, P7). However, after becoming users of DouYin or KuaiShou, the favoured conversation topic was how to perform better on the platforms. For example, P10, who works in a student dormitory, said:*"Before I used DouYin, I gossiped with others every day. This was nonsense and boring for me. After using DouYin, we could talk about the skills of performance."*

5.3.2 Ageing with Self-reflection. Besides benefits brought by skill learning, we found that short video platforms also helped older adults adopt positive attitude towards later life. As mentioned, others' feedback and expectation offered our participants a lot of power to continue practising and

[9] A popular application for instant message sending in China.

improving. Moreover, we found that others' feedback may also help to shape their life attitude. For example, P1 told us she realized that she was not happy when receiving others' comment that why she looked so sad and she began to post happier videos and learned to become positive since she thought it was better to spread happiness on the platform.

Moreover, many reported that many content posted by other users empowered them a lot. Similar to younger users, our older adults were also engaged by videos of "Positive Energy" on the platform [45] and felt the kindness of humanity. With the rich UGV made available online, they could find videos that could easily resonate with their own lives and get touched and impacted. Take P1 and P10 for example,

> *"There are many meaningful idioms on the platform. I didn't understand these things when I was young. I think DouYin should make an effort to promote these words, which are good for people." (P1)*

> *"My son is disabled. I often quarrel with him because my son is depressed. However, I saw many positive children on DouYin, who are also physically disabled. And people around them take care of them. After watching these videos, I also feel my responsibility to take good care of my son, and I should be more positive." (P10)*

Nevertheless, among videos echoing older adults' feelings, we found a tricky kind of video called as "Toxic Chicken Soup for the Soul" ("毒鸡汤"). This kind of videos is generally featured by extremely emotional and seemingly reasonable expressions (e.g. "Girls/Women, men unwilling to give you money definitely don't love you."). While regarded as toxic by some younger user, some of our participants were also moved by these content and agreed with the notion.

Besides, increasing social interaction with others also brought up opportunities for understanding of the value of life. For example, P5 regards the feedback from his followers as the most valuable gaining of video streaming,

> *"The feedback received on the platform makes me happy because this means that people regard my performances as good. Even if people come to talk with me, not because of my performance, I am also happy, since this also shows respect to me."*

Similarly, P6 also felt that sharing is possibly the best way of gaining. Through offering support to others, P6 felt that he had made a real impact, which made him feel his retired life meaningful:

> *"I've influenced many people, which makes me feel responsible. For example, I often talked with one guy from Shanghai during live-streaming and I helped to solve many of his worries toward life. Now we have become friends and he sometimes would send me gifts, which made me feel very good."*

6 DISCUSSION

To our knowledge, this study is the first work of how older adults use short video platforms, thus filling the research gap and spotlighting the unrepresented yet important population on short video platforms. Surprisingly, short video platforms have attracted a large number of older adults to contribute their own contents and share with others, which is largely contradict to the stereotype of ageing [82]. In this section, we will mainly focus on the insight achieved from our findings for how to better support older adults' active use of technology, especially their online creation and sharing. Admittedly, technology has its limits. In this study, we mainly focus on the perspective of older adults who actively engaged with short video platforms and how they perceived these platforms in their own words. Future research may be considered to see how general older populations used and perceived short video platforms, to gain a more comprehensive understanding of senior users and these platforms.

6.1 Providing a Low-barrier yet Fruitful Start for Older Adults

We found that one of the main advantages of short video platforms is that they provided a low-barrier yet fruitful start for older adults, which offers our older adults both enjoyment and confidence for the start of use. According to our findings, this is largely supported by the media form (i.e., "video"), as well as the diverse content in both varieties and qualities. Specifically, video, as an accessible yet dynamic form of media, helps to lower the bar for participation while supporting rich and meaningful expression. The content diversity in quality also helps to lower the bar for creation, while diversity in variety contributes to the fruitfulness as older adults could easily find content of their interests and personal relevance.

6.1.1 "Video" as an Accessible yet Rich Media for Older Adults. Obviously, the media form (i.e., video) marks the main difference between short video platforms and other traditional text/image-based online communities. In this study, we revealed the main value of "video" as an accessible yet rich media form for older adults. That is to say, video provides a low entry bar for older adults to easily participate, but it is also rich in expression and meaning. While past work shown that even the elderly people who are members of online communities rarely publish and share pictures, or watch and listen to others' video and photos due to difficult use of UGC applications [38], we found that video as a media form could be a good way of expression for older adults.

Video provides a much lower bar for content creation than text. Many of our participants feel reluctant to share on Weibo, a micro-blogging application, largely because the use of Weibo needs typing, which is not friendly to many older adults [60]. For illiterate older adults or those without high education level, things may get even worse. It is difficult for them to use a text-based application such as Weibo since they cannot read and write. Therefore, video, as a direct and rich form of media, is especially meaningful for this group of older users. In our study, many of our participants are from rural area and did not receive much education, but they are also able to participate in video creation and receive others' "Likes". This shows us a positive picture of engaging this large group of people into online creation and sharing, who are generally excluded from digital technologies [22] and creative activities.

While accessible, video is a rich form of media, which enable our older adults to show all aspects of themselves. Many of our participants appreciated short video platforms as these platforms provide a stage for them to show all of their talents, such as musical instrument playing, singing and dancing. These performances are hard to be delivered by static media forms such as text and image. Some research has already indicated that image-based social media platforms such as Instagram contributes more to users' satisfaction with life than text-based ones such as Twitter [57], possibly because images could facilitate social presence [71]. Goh et al. similarly found that photos were the medium of choice for individuals to share, because they quickly got the job done in terms of communicating feelings or situations [29]. Videos, as a much richer form of media than image, may further facilitate older adults' expression and performance by representing more human characteristics such as voice, language, and personality [71].

What's more, unlike static forms of media, video creation usually requires many physical activities, which is important for older adults to keep active after retirement [20]. As our findings show, to improve their performance skills, older adults usually spend much time practising. Some would even go out to have a better view for their followers. However, we also notice that short video platforms may replace many offline activities of older adults. For example, some of our participants gave up or decrease the frequency of public square dancing, karaoke and Mahjong, and some also usually lie on the bed when watching videos on the platform. We guess this may be even worse for passive older users who only watch videos on the platform. Future research may be needed to see the relation between physical activity and short video platform use.

Another characteristic of videos on short video platforms worth noting is the short format – most videos on short video platforms are within minutes. While past research noted that older adults preferred long forms of expression because they value the depth of thought [11], our study shows short videos as a short yet rich form of media for older adults. The short length means a lot for our older users, since they usually have limited physical capacity. This is shown more clearly in the reason reported by older adults who choose not to hold live-streaming. The main concern of them is exactly the cost of energy and time. The short length may also facilitate older adults' expression as they don't have to consider the structure of video, which is reported as an important factor in long UGV or live-streaming [68].

6.1.2 Content Diversity as a Strategy to Lower the Entry Bar while Adding Fun. Another factor contributing to the low entry bar and fruitfulness of short video platforms is the diverse content in both varieties and qualities. Notably, some participants with assumed fear of technology also tried to use the application with their interests and encouragement of their family members, and they also felt comfortable in the further use. This shows the importance of stimulating older adults' interests and lowering the entry bar for older adults to gain both interests and confidence in the initial use of new technologies. Past research on Korean senior UGV creators already noted the importance of perceived benefit, ease of participation and enjoyment based on survey [62]. Our work provides a more detailed description of older adults' initial perception of UGV applications, and what matters for them at the start of use.

For our participants, their decision to start using the application is largely attributed to the videos they saw when first opening the application. All of them are soon attracted and they found that they are also able to create similar work since the quality of videos recommended are not so high. These findings are highly consistent with past findings that older adults are resistant to social media mainly because they perceived social media as superficial and meaningless [41]. Perceived difficulty is also a large barrier for them to use UGC applications [38]. However, they may change such negative impression and become willing to share content relevant to their life after proper educational strategies [87]. Indeed, many older adults, including our participants, have needs to express themselves online [11], especially considering that older adults have a large amount of leisure time after retirement and it is important for them to develop their interests and establish later-life identity [11]. Older adults also have the ability to engage in creative activities [61]. However, their initial perceptions of new technology may be a large barrier for them to adopt new technologies and UGC creation. Therefore, it is important to make older adults feel relevant and confident from the beginning of and even before the use if we want to engage older adults into active use of technology.

Notably, short video platforms present us a successful case by providing a fruitful start and a low entry bar that engage older adults into active use of new technologies for creation without any extra educational strategies. Our work then enrich the current research by a detailed description that how technical features, such as recommendation strategy in this case, may influence older adults' initial perception of new technology. In our work, we mainly reveal the value of the decentralization policy of recommendation algorithm to older adults. With the decentralization policy, older adults have gained initial motivation and confidence as the videos on the homepage are diverse in both variety and quality, which made them easily find videos echoing their hearts and feel the bar for participation comfortable. By contrast, Weibo, as a negative case, though also supporting video sharing and live-streaming, mainly promotes content from celebrity and recognized account, which are perceived as far away our participants' life and made our older adults feel resistant to use. Similarly, YY and WeSing, though similarly driven by UGC, failed to attract our older adults since the content is only about entertainment and seems monotonous.

On the other hand, while diversity in recommendation may bring about these benefits, we admit that it may also increase the risk of recommending negative and controversial content. In this work, except for some erotic videos, our participants are generally left with a good impression since the platforms have already strictly regulated content posted. However, we also noticed that many videos online are controversial and cannot be simply defined as negative. For example, we found a a tricky kind of content, which is usually featured by extremely emotional and seemingly reasonable expression and generally called as "Toxic Chicken Soup for the Soul" by younger users [6]. We suggest designers should pay attention to such kind of content. Large-scale annotation and labeling may be effective to make users cautious [26], and diversification of comment may also enable deeper discussion about such controversial issues [27].

6.2 Creating a Rewarding Experience for Older Adults to Continue

While the low entry bar and personal interest lay the foundation for our older adults to start UGV creation, finally it is interaction with people and social support received on the platform made them stay on the platform and continue creating and sharing. That is, short video platforms have created a stage socially constructed and sustained by the users for older adults. Similar findings could be found that it is difficult for older adults to engage in creation and sharing without responsive audience [61]. However, the inner mechanism that how active older adults interact with broad users on UGC platforms is still unclear.

For older adults, extrinsic motivation such as monetary incentives is not so important for older adults as the case of younger users [80], and they usually put much more value in others' appreciation and encouragement. The positive feedback and respect from others are what ensured the continuing engagement of older users. Here, Weibo is again presented as a negative case. Though some of our participants used to use Weibo for content creation, all of them finally gave up because there is no response and they could not see the meaning of sharing on that platform. Therefore, to engage older adults in continuing creation and sharing, in-time response and appreciation from others is highly important.

What's more, our participants not only appreciate others' response and feedback but also want to benefit others. That is to say, reciprocity is highly valued by our older adults. Specifically, our older adults often consider the altruistic value of their content (e.g. whether they could bring happiness to others), and they enjoy posting performance-related videos rather than personal life because they regarded showing their personal life as meaningless to others. This finding echoes well with past finding that older adults who actively engage in blogging perceive their content posted as valuable to others [11]. This is very different from younger UGC creators, who mainly create UGV for their own incentives such as information seeking and ego-defensive goals [39]. Also, altruism and perceived reciprocal benefits are not significant factors for general adults' continuance motivation to use YouTube [18].

Another interesting finding worth noting in our study is that older adults are highly willing to show their encouragement and support to others because they knew the importance of support for UGC creators. The motivation of our older adults' for sending "Like" is very different from younger users' – for elderly users, as long as they feel the efforts paid in the video, they will send "Like" to show their support and encouragement, regardless of the quality of the video. Some even sends "Like" every video he viewed. This is highly different from younger users, who mainly sent "Likes" out of hedonic and utilitarian motivations [19]. As such, a positive cycle has been made among older adults and other users – older adults show their support to others and meanwhile supported by their viewers, thus making the sharing experience rewarding and satisfactory.

Again, the decentralization policy plays an important role here to increase interaction between our participants and other like-minded users. As mentioned, due to decentralization, the algorithm

will recommend videos of various qualities, and this means that work from our older adults also has a certain degree of chance to be viewed by others on short video platforms. This is why our participants could receive feedback on short video platforms so fast, while hard to get response on Weibo, since Weibo mainly promotes content from celebrity and recognized account. However, on the other hand, it should be admitted that such policy may increase the concern about privacy issues. In past work, older adults show general concern with privacy issues on social media [41]. In our study, all of our participants choose to believe the kindness of most users on the platform, though some had some negative experience such as online fraud. This may be due to the Chinese context of our study, where people show less concern with privacy than users from Western countries [81]. Considering this, although distributing older adults' work to large-scale audience may increase older adults' motivation for sharing, we suggest designers may consider making the distribution process more transparent to our older adults and adding features to let senior users to choose who to share and the size of audience.

6.3 Offering Older Adults a Comfortable Learning Curve for Growth

While short video platforms provide a low entry bar for start, they provide much learning opportunities in the further use, to which our older adults feel grateful a lot. While many researchers worked hard to make technology accessible to older adults, the case of short video platforms presents us a new possibility that older adults could adopt and learn to use new technology with little help from outside if they are motivated. What's more, the learning opportunities brought up may further increase the satisfaction of older adults since they could feel their growth during the use, which may further add value to their retired life.

Past work noted that older adults enjoy watching meaningful videos on YouTube that they could learn something from (e.g. golf) [63]. Our study further show that, not limited to videos watching, older adults could learn various things during the learning process of the use of the application. While accessible, the videos on short video platform provided much room for older adults to adapt and show their creativity. As shown, our participants are not satisfied with the basic use of the application. Our older adults are also actively considering how to improve their video quality, how to add fun for their viewers, how to better communicate with other users, and they even reflect over their life attitude during the use of short video platforms. As such, the learning on short video platforms could cover every aspect of our participants' life, which is far beyond simplistic information browsing and exchanging in traditional online communities [43, 52].

In addition, learning to use the application may contribute to older adults' social integration after retirement. Some past work has already noted that special technology for older adults may not be a good idea since it may foster social exclusion and senior users may avoid using such technologies [64]. Similar results are also reflected in our study. As our work shows, learning to use the application creates many opportunities for interactions, especially substantial interactions valued by older adults instead of meaningless talking and gossip, which is based on the fact that they are using the same application with others, rather than application specially for older adults. When facing difficulties of use, our older adults would leverage all channels to ask for help, including direct message sending, chat groups and asking family members or offline friends who are also using short video platforms. Learning, then becomes a talking ticket for older adults to socialize, including online and offline, inter-generational and intra-generational. Moreover, our older adults could even teach others how to use specific features and improve the quality of videos. Thus, learning and teaching how to use the application then become an important part of their retired life. The difficulty in use could also be moderated by such learning-oriented socialization.

Here, we are not to argue that accessibility is not important for design for ageing. Many studies have already revealed the importance of perceived ease of use for older adults, and the elderly

generally expressed their worries about learning new technologies [38]. Indeed, while short video platforms need users to learn a lot if users want to make high-quality videos, they provide a comfortable start threshold with clear interface design and easy-to-use features to facilitate video making. Also, the design putting recommendation page as homepage largely facilitates the information seeking, which offers our participants much confidence to use. However, the accessibility is not the whole story. Many our participants have also faced many difficulties during the use, but they could overcome them by themselves as they are motivated. For example, many our participants did not know how to make some videos using special effect filtering, but they would send direct messages or ask their family members or friends, and then learn to make it. Also, some older streamers felt nervous when they started live-streaming, however, with their motivations, they finally overcome all the difficulties by practising.

Combining our study and current literature, we argue that the comfortable learning curve for older adults is highly important when design ICTs for older adults. We suggest designers should leave room for older adults to learn and grow while paying efforts on accessibility. This may be beneficial for older adults to use technology in a long term. In addition, designers could pay more attention on motivating older adults into active learning, rather than design a specific training course [31] or special version (usually with lower functionality) [64] to facilitate their use. For example, in the case of short video platforms, it is others' appreciation and response that offer the main motivation for our older adults to learn to use the application. Designers may also consider including older adults into the co-design of technology [40].

6.4 Design towards Active Ageing

Finally, we want to argue that the use of short video platforms among older adults echoes well with the concept of "active ageing", which may help designers to think about what is a better way of ageing and how to leverage technology to support ageing better. Active ageing, defined as "the process of optimizing opportunities for health, participation and security in order to enhance quality of life as people age" by World Health Organization (WHO) [53], has been adopted as a global strategy in ageing policies, practices and research over the last decade [50, 54, 79] for its inclusiveness since it adopts a wide definition for health – a state of physical, mental, and social well-being [91]. Specifically, an active ageing policy does not require complete health, but should enable everyone to age as well as they can [91].

According to our findings, the three dimensions of active ageing (i.e., physical, social, mental) are well reflected in the use of short video platforms among our participants. As mentioned, video creation requires many physical activities, which may avoid older adults from keeping sedentary and thus may contribute to their physical health. In terms of social well-being, many of our participants have built new social connections with others on the platform through encouragement and advice exchanging, and pre-built social ties may also be strengthened during the learning of the use. More importantly, such social interaction is perceived as meaningful by older adults, rather than superficial expressions and showing off on social media [41]. Mentally, many of our participants become willing to learn new skills and feel empowered during the use of short video platforms. More importantly, to be an active learner is beneficial for their sustainable wellness by contributing to psychological qualities including "self-esteem", "self-efficacy" and "resilience" [25]. To testify the relationship between the use of short video platform and related benefits for older adults, quantitative research could be considered for future research.

As such, the use of short video platforms is far beyond the basic and mundane use of ICTs [66, 78], covering all aspects of older adults' retired life and helping older adults ageing well in a long run. In this sense, older adults' active use of short video platforms presents us a new positive narrative of older adults and ICTs. In past research, most work focused on the downside of ageing

(e.g. reduced physical functioning), treating older adults as the declined youth and technology as the assistive solution to compensate for specific loss [77]. However, we found that technology could be leveraged to serve multiple goals together based on the concept of "active ageing". For the case of short video platforms, though originally designed for video creation and sharing, they could be leveraged to support older adults' physical activities, learning-oriented socialization and lifelong learning, which, as a whole, may contribute to an active and meaningful ageing experience. While previous research often treats ageing as a state where older adults' capability and potential are degraded, mainly focusing on accessibility and designing senior friendly technology for specific goals [40], we suggest designers should take a holistic view about ageing, treat ageing as a process and consider long-term benefits for older adults who can still participate in active activities, such as performance sharing, gaming [76] and technology hacking [70] etc. As such, active ageing could be enhanced in a sustainable manner.

7 CONCLUSION

We have presented findings from a qualitative study to understand how older adults use short video platforms and how these platforms support older adults' active use of technology. We found that short video platforms offer an accessible yet fruitful and encouraging stage for older adults to start and continue. With such a friendly learning curve, older adults would feel comfortable with the initial use, and feel their growth in the further use. As such, an active use of technology is made possible – older adults show their talents, interact with like-minded peers, learn new skills, and even reflect over their life attitude. These, as a whole, help to promote active ageing and lead to a meaningful ageing experience. Our findings provide important insights for how to engage older adults into active use of technology and what might be a better way to support ageing as a process with technologies.

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